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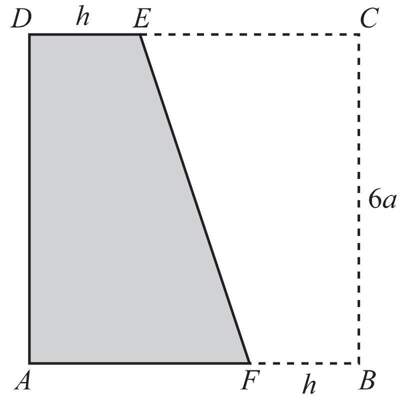

$ABCD$ is a uniform square lamina of side $6a$. Points E and F are on DC and AB respectively and are such that $DE = FB = h$. The quadrilateral $BCEF$ is removed from the square lamina (see diagram).

(a) Show that the distance of the centre of mass of the resulting lamina $AFED$ from AD is $\dfrac{h^2 - 6ah + 36a^2}{18a}$ and find a corresponding expression for the distance of the centre of mass from AB. [5]

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When the lamina $AFED$ is suspended from the point D, the edge DA makes an angle θ with the downward vertical, where $\tan\theta = \frac{7}{15}$.

(b) Find, in terms of a, the two possible values of h. [3]

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